

EDUCATION

- **2020 - 2023:** Doctor of Philosophy, Thesis Title: “Knowledge Distillation and Continual Learning for Optimized Deep Neural Networks” - University of Wollongong.
- **2017-2019:** Bachelor of Computer Science - GPA 9/10 - University of Wollongong.

ACHIEVEMENTS

- **2023:** \$40k research grant [1*] on tumor segmentation from Channel 7 Children’s Research Foundation as *Chief Investigator* ([media link](#)).
- **2022:** \$50k research grant [4*] on scene segmentation for road conditioning assessment from QLD Department of Transport and Main Roads as Co-Investigator.
- **2021-2023:** University Postgraduate Award - Ph.D Scholarship from University of Wollongong.
- **2020:** Best research paper award in IEEE SmartIoT Conference.
- **2017-2019:** Dean’s Merit Award (annually for the top 5% in the faculty) - University of Wollongong.

AI/ML EXPERTISE

- Generative AI [2], multi-modal model [1], [3], [7], prompt tuning [4], and explainable AI [13], [3].
- Vision Transformer [6], knowledge distillation for continual learning [5], and model compression [8],[9].
- Natural language understanding [11].
- Applied AI: tumour segmentation, medical image synthesis, and diagnosis [1*,2*,3*,5*].

PROFESSIONAL EXPERIENCE

Google

Research Scientist

Sydney
Oct 2025-Now

- Research on multimodal LLMs to push fine-grained visual perceptions.
- Develop foundational and multimodal LLMs for biometrics to ensure sustainability, i.e., lower re-training and data collecting cycles.

Australian Institute for Machine Learning, Adelaide University

Postdoctoral Research Fellow

Adelaide
16 Nov 2022 - Aug 2025

- Develop state-of-the-art vision-language models, outperforming radiologists on X-ray disease classification by 6%.
Boost 30% radiologist report writing time - reduce \$3.8K daily cost of radiologists.
Published at **CVPR2024** [1] (top-1 global AI conference) with 70-star open-source model on [Github](#).
- Develop and deliver tumor segmentation pipeline for patient treatment planning at South Australian hospitals. Won three nationally competitive research grants [1*,2*,3*] from clinical research centers, totaling \$127k.
- Develop LLM with prompt engineering to combat hallucinations for trustworthy medical applications. Boost disease diagnosis accuracy of LLMs by 5%.
Collaborate with clinicians to define machine learning scopes. Work with medical professionals to collect, annotate data, and evaluate LLM hallucinations for high-stake disease diagnosis.
- Develop generative models to generate CT images from MRI, avoiding patients’ radioactive exposures. Minimized harmful CT scans for 72% of pediatric patient intakes.
Publish advanced generative models at MICCAI2024 [2], and MICCAI2023 [7], top conference in medical imaging ([Github](#) repo here).
- Mentorship and leadership: Supervise 2 PhD and 3 postgraduate students on deep learning research. Publish a Vision Transformer model at top IJCV journal (Impact Factor 11.6) [6] and explainable AI at CVPR2024 [13].

University of Wollongong

Machine Learning Engineer

Wollongong
Nov 2021 - Nov 2022

- Develop machine learning models for automatic bird counting, fire detection and fire risk assessment models for the AI-IoT environmental management system.

- Deploy AI models on edge using Azure IoT Edge.
- Collaborate with ecologists in Vietnam's Tram Chim National Park to design and deploy a scalable software pipeline tailored for field use.
- Continuous model development: monitored and retrained models to adapt to evolving environmental data, ensuring ongoing accuracy and relevance.
- High-impact recognition: receive endorsement from Australian ambassador and Vietnamese government for its impact in protecting forest and biodiversity: [media link](#), [project website](#).

University of Wollongong

AI/ML Research Assistant

Wollongong

Feb 2022 - Nov 2022

- Develop detection models to assess road conditioning for Department of Transport and Main Roads.
- Build a mapping of 10 road defects and 11 road types on 4,109 km of state-strategic road.
- Awarded a grant [4*] worth \$50k from Department of Transport and Main Roads, Queensland, Australia.

ASX

Data Scientist

Sydney

Sep 2019 - Feb 2020

- Develop computer vision model to extract texts and tables from PDF files.
- Build text mining tools to extract entities and their relationship in PDF files.
- Reduced annual data entry costs by \$40k, demonstrating measurable business impact through automation.

AI/ML RESEARCH PROJECTS

Multi-modal LLM

Postdoc Research Scientist, AIML, University of Adelaide

Aug 2023-Now

- Develop vision LLM for high-stake disease diagnosis. Publish at top-tier global conference CVPR2024 [1]
- Work with clinicians, and customers to collect, and annotate data. Design end-to-end pipeline for training and evaluating LLM and vision-language models; align AI solutions with business needs.
- Enhance LLM's reasoning for clinical diagnosis using retrieval-augmented generation (RAG) and chain-of-thought prompting. Publish at **EMNLP**, US, 2024 [4] (top-2 conference in NLP).

Generative model

Research Scientist, AIML, University of Adelaide

Jan 2023- Now

- Develop Transformer-based generative AI for image generation.
- Resolve structural deformation of current generative models by integrating anatomical priors. Publish MICCAI 2024 (Top 11% - early accept) [2], and MICCAI 2023 [7].

Knowledge distillation for continuous training and model compression

Ph.D, University of Wollongong

Nov 2021-Nov 2022

- Develop advanced continual learning frameworks to enable continuous update and prevent forgetting. Publish CVPR 2022 [5].
- Develop compression algorithms for deployable segmentation models in biosecurity scanning applications. Publish Neurocomputing 2022 [10].

AI-IoT for smart appliances

IoT, ML Engineer

Aug 2020 - Feb 2021

- Develop an AI-IoT system for smart appliances. Develop ML models classify signals to optimize energy savings and understand consumers' behaviors.
 - Deploy models on smart IoT sensors for monitoring energy loads.
- My system achieves Best Paper Award at an international conference, IEEE SmartIoT.

Aspect-based sentiment analysis on product reviews

B.CompSc, University of Wollongong

Jun 2019 - Dec 2019

- Discover the lack of syntactical understanding of language models, and develop syntax-enhanced LMs. Publish **ACL** 2020 [11], top-1 global conference in NLP; attract 200+ citations ([link](#)), showing NLP industry and research impacts.

GRANTS

- [1*] Channel 7 Children's Research Foundation, "AI-assisted Contouring of Sarcomas to Improve Safety and Efficiency of Proton Therapy", **Chief Investigator**, \$39,570, 2023.
- [2*] NeuroSurgical Research Foundation, "Paediatric Radiation Therapy with AI-based Segmentation for Enhanced Treatment Planning", *Associate Investigator*, \$47,790, 2023.
- [3*] ANZSA, "Improving the Radiological Evaluation of Bone and Soft Tissue Sarcomas with Advanced Diffusion Imaging and Artificial Intelligence", *Associate Investigator*, \$40,000, 2023.
- [4*] TMR Spatial Labs Program, "Developing Deep Learning Tools for Road Information Extraction from Georeferenced High-resolution Imagery", *Co-Investigator*, \$50,000, 2022.

Under review:

- [5*] Bloom Research Program, "BRinging Artificial Intelligence iNto pediatric oncology: the myBRAIN translational trial project", *Co-Investigator*, \$2M, 2024.

PUBLICATIONS

- [1] **V. M. H. Phan**, et al., "Decomposing Disease Descriptions for Enhanced Pathology Detection: A Multi-Aspect Vision-Language Pre-training Framework," *IEEE Conference on Computer Vision and Pattern Recognition*, 2024, Accepted (Top-1 conference in AI, H-index=422); [Presentation link](#).
- [2] **V. M. H. Phan**, et al., "Structural Attention: Rethinking Transformer for Unpaired Medical Image Synthesis", *International Conference on Medical Image Computing and Computer-Assisted Intervention*, Early Accept (Top 11%), 2024. [Paper link](#)
- [3] F. C. Townim, **V. M. H. Phan**, et al., "AdaCBM: An Adaptive Concept Bottleneck Model for Explainable and Accurate Diagnosis", *International Conference on Medical Image Computing and Computer-Assisted Intervention*, 2024.
- [4] T. D. Nguyen, T. T. Huynh, **V. M. H. Phan**, et al., "CARER - ClinicAI Reasoning-Enhanced Representation for Temporal Health Risk Prediction", *EMNLP*, 2024 (Top 2, conference in NLP, H-index=193). [Paper link](#).
- [5] **M. H. Phan**, T. Ta, S. L. Phung, L. Tran-Thanh, A. Bouzerdoun, "Class Similarity Weighted Knowledge Distillation for Continual Semantic Segmentation," *IEEE Conference on Computer Vision and Pattern Recognition*, 2022 (H-index=422).
- [6] B. Zhang*, L. Liu*, **M. H. Phan***, et al., "SegViTv2: Exploring Efficient and Continual Semantic Segmentation with Plain Vision Transformers", *International Journal of Computer Vision*, 2023 (IF=19.5). *Equal contribution
- [7] **M. H. Phan**, et al., "Structure-Preserving Synthesis: MaskGAN for Unpaired MR-CT Translation", *International Conference on Medical Image Computing and Computer-Assisted Intervention*, 2023 (H-index=89).
- [8] J. Yuan*, **M. H. Phan***, et al., "FAKD: Feature Augmented Knowledge Distillation for Semantic Segmentation", *IEEE Winter Application on Applications of Computer Vision*, 2024 (H-index=95). *Equal contribution.
- [9] L. Liu, Z. Wang, **M. H. Phan**, et al., "BPKD: Boundary Privileged Knowledge Distillation For Semantic Segmentation", *IEEE Winter Application on Applications of Computer Vision*, 2023 (H-index=95).
- [10] **M. H. Phan**, T. S. L. Phung, K. Luu, A. Bouzerdoun, "Efficient Hyperspectral Image Segmentation for Biosecurity Scanning using Knowledge Distillation from Multi-head Teacher," *Neurocomputing*, 2022 (IF=6.0).
- [11] **M. H. Phan**, P. Ogunbona, "Modelling context and syntactical features for aspect-based sentiment analysis," in *Association for Computational Linguistics*, 2020 (Top-1 conference in NLP, H-index=215). [Presentation link](#).
- [12] V. K. Nguyen, **M. H. Phan**, "A hybrid approach for intrusive appliance load monitoring in smart home", in *IEEE International Conference on Smart Internet of Things (SmartIoT)*, 2020, *Best Paper Award*.
- [13] F. C. Townim, K. Liao, **V. M. H. Phan**, et al., "CAPE: CAM as a Probabilistic Ensemble for Enhanced DNN Interpretation," *IEEE Conference on Computer Vision and Pattern Recognition*, 2024, (Top-1 conference in AI, H-index=422).

REFEREES

- Prof. Son Lam Phung. Email: phung@uow.edu.au.
- Dr Yuankai Qi. Email: yuankai.qi@mq.edu.au.